

ANDCS × MIT HACKING MEDICINE · SUSTAINABLE CARE · PARIS 2026

Dialysis has a hidden water vulnerability.

EcoDialysis turns wasted dialysis water into a managed resilience asset — measurable, redesigned, certified.

270 L rejected per session 15,600 sessions / yr / center ~4,200 m³ wasted / yr / center

Strategic agenda — from hidden waste to certified water resilience

Six moves. Each one is also an EcoDialysis revenue line.

1 · SEE THE PROBLEM

Make a hospital's hidden dialysis water flow visible at session level.

2 · CHOOSE THE RIGHT REUSE

Green / orange / red map of every non-clinical reuse route.

3 · FILE THE DOSSIER

EICH-compliant ARS / préfecture safety case per site.

4 · IMPLEMENT & METER

4-point metering, separated network, anti-backflow, carnet sanitaire.

5 · VERIFY SAVINGS

SaaS dashboard, monthly verification, KPI proofs to ARS and finance.

6 · CERTIFY & SCALE

EcoDialysis label → GHT / private-network rollout → European expansion.

Adaptation first, mitigation second. **Pch** **S26** **Gov**

Why water, why now — hospitals enter a water-constrained decade

THE NEW OPERATING REALITY

- **Scarcity:** France basins facing structural deficit (Loire, Adour-Garonne, PACA).
- **Regulation:** Décret n°2024-796 (12 July 2024, in force 1 Sept 2024) opens non-clinical reuse pathway.
- **Tariffs:** water + sewer indexation hitting hospital opex; sewer often > potable.
- **Continuity:** droughts and restrictions threaten technical functions in summer.
- **Reporting:** Planification écologique du système de santé requires per-establishment water plans. **Gov** **TSP** **Ird**

Water will become an **operating constraint**, not a sustainability metric.

WHY HOSPITALS CAN'T WAIT

- Healthcare = **>8% of national GHG**, ~50 MtCO₂e (offer of care 45%, products 55%).
Gov **HCW**
- Care offer = buildings, energy, **water**, waste, transport.
- Dialysis is the most water-intensive recurring hospital workflow.
- Water adaptation pays back in opex **before** carbon counts.

Why dialysis is the right first hospital department

MEASURABLE

Closed technical loop. Every session has a known feedwater, reject volume, energy draw. ~**400–500 L** feedwater, **50–70%** reject. **Per** **Gem** **Ope**

RECURRING

~156 sessions / patient / year. 100-patient center = **15,600 sessions / yr**. Predictable savings.

TECHNICALLY ACTIONABLE

RO reject = controlled technical stream, not patient wastewater. Reuse routes already proven internationally. **Ope** **Gem**

CLINICALLY BOUNDED

No patient contact reuse. Never back to dialysis-water production. Never spent dialysate. Boundary is the wedge of trust.

POLICY-ALIGNED

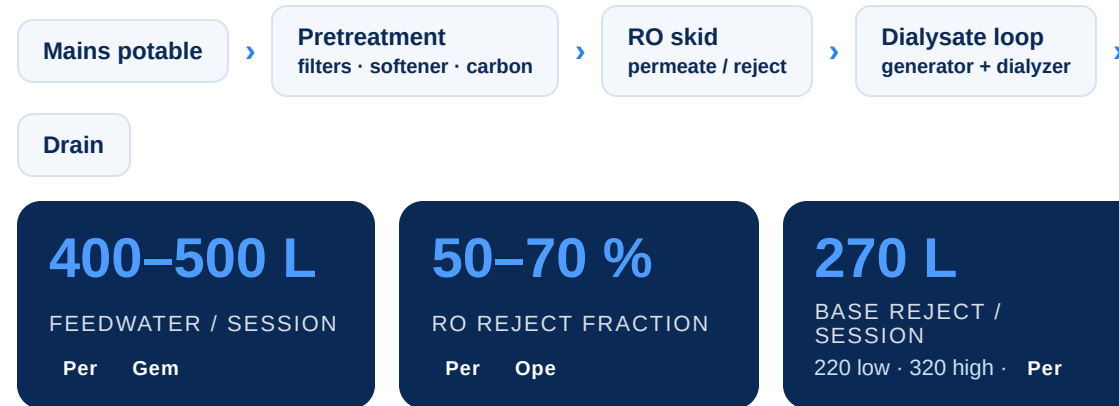
SFNDT Guide des bonnes pratiques 2026 explicitly covers RO-reject reuse. **S26** **S23**

STRATEGIC

Win dialysis first → expand to sterilization, laundry, kitchens, technical loops.

What exactly is wasted — the dialysis water flow

ONE SESSION, SIMPLIFIED



THE HIDDEN THREE STREAMS

- **RO concentrate (reject):** clean, low-salt, controlled. Highest reuse value.
- **Disinfection & rinse water:** intermittent, contains residual chemicals — reuse only after dilution / neutralization.
- **Spent dialysate:** patient-contact — **never** reused. **RED**

HIDDEN ENERGY TOO

Older Australian RO: **548 L + 7.2 kWh** / treatment. Modern RO: **357 L + 3.1 kWh**. **Ope**
East Kent heat exchangers: **0.86 kWh/treatment** recoverable, **27,905 kWh/yr** projected.

Part 2 — Measurement & Reuse

If you cannot measure the water, you cannot redesign it.

If you cannot redesign it, you cannot certify it.

How the water is measured — 4-point metering architecture

THE FOUR METERS

- **M1 · Mains-in** — total feedwater entering the dialysis water plant.
- **M2 · RO permeate** — water actually delivered to dialysate loop.
- **M3 · RO reject** — controlled technical stream candidate for reuse.
- **M4 · Reuse delivery** — water actually substituted in toilets / cooling / cleaning.

PLUS CONTEXT CHANNELS

- Conductivity on M3 → routing logic (high salt → drain, low salt → reuse). **Gem**
- Pressure + flow on reuse loop → leak and stagnation detection.
- Energy submeter on RO + heat-exchanger (optional).
- Time-stamped to session log → L/session benchmark.

WHY MEASUREMENT-FIRST WINS

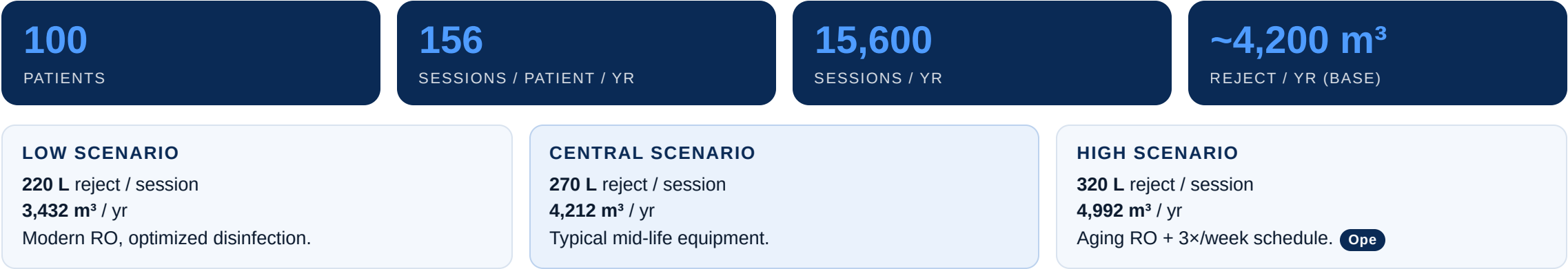
SFNDT 2023 and SFNDT 2026 both place **measurement-first** at the heart of green nephrology. **s23 s26**
TSP Pathway methodology: objective, perimeter, transparent assumptions, uncertainty studies. **TSP**
No measurement → no ARS dossier, no CEE, no savings claim.

FROM METER TO EVIDENCE

- Baseline 4–6 weeks before any intervention.
- Continuous logging post-intervention.
- Monthly verification report to direction, technique, hygiene.
- Quarterly KPI bundle to ARS & finance.

Reference center model — 100 patients, 15,600 sessions

A standard mid-size French in-center / UDM unit. The math hospitals can verify against their own bills. Per



Water-only carbon caveat: ~1.0–1.1 tCO₂e / yr for the 4,200 m³ band — secondary to opex resilience. Per TSP

What can be reused — the green / orange / red map

GREEN — REUSE NOW

- Toilet flushing in non-clinical zones
- Floor & outdoor cleaning
- HVAC cooling-tower make-up (with treatment)
- Garden / outdoor watering
- Technical washing (carts, bins)

Evidence: Canterbury, Bristol, Ashford, ANZSN.

[Ope](#)

[Gem](#)

ORANGE — SITE-SPECIFIC

- Laundry pre-wash
- Sterilization feed (non-final-rinse)
- Boiler / cooling make-up with extra polishing
- Pretreatment recycle (recovery RO) — vendor-specific
- Disinfection rinse reuse — only after neutralization

Caveat: requires dossier, local water quality, vendor sign-off.

RED — NEVER REUSE

- Spent dialysate (patient contact)
- Any return to dialysis-water production
- Drinking / food-prep water
- Patient-contact equipment final rinse
- Endoscope / surgical-instrument final rinse

Non-negotiable. The safety boundary is the wedge of trust.

The question is **not** whether water can be reused.

The question is **which water, where, under which legal and safety conditions.**

Current law vs new EICH framework

BEFORE — PRE 1 SEPT 2024

- No general non-potable reuse framework in healthcare.
- Ad-hoc derogations from ARS / DDETSPP.
- Reuse projects stalled in legal uncertainty.

AFTER — DÉCRET N°2024-796, IN FORCE 1 SEPT 2024

EICH = Eaux Impropres à la Consommation Humaine (non-potable waters) framework for sensitive establishments including healthcare. **Gov** **S26**

- Permits non-clinical reuse with site-specific authorization.
- Requires ARS / préfecture safety case.
- Separate network, anti-backflow, signage, carnet sanitaire.
- Monitoring & emergency stop required.
- Defines authorized uses, not a blanket licence.

WHAT EICH DOES NOT DO

- Does not authorize any patient-contact reuse.
- Does not pre-approve any site — every dossier is local.
- Does not waive hygiene committee (CLIN) sign-off.
- Does not guarantee tariff savings — local sewer / potable structure decides.

EICH opens the door.

EcoDialysis builds the dossier.

Caveat: exact ARS pathway and monitoring frequency vary by basin and establishment type. Confirm at each filing. See Appendix H. **Gov**

What an ARS / préfecture dossier must contain

MANDATORY TECHNICAL CONTENT

- Source description: RO reject characterization (conductivity, microbio, chemicals).
- Use description: target uses, volumes, locations.
- Separated network: schematic, materials, anti-backflow devices (EA / EB / AA).
- Storage: volume, residence time, materials, anti-stagnation logic.
- Treatment if any: UV-C, filtration, disinfection.
- Monitoring plan: parameters, frequency, alarms.
- Emergency stop & fallback to potable.
- Carnet sanitaire entry and responsibilities.

MANDATORY GOVERNANCE CONTENT

- Risk analysis (HACCP-like) per use.
- Sign-off chain: direction, technique, hygiène (CLIN), médical, qualité.
- Staff training plan + signage program.
- Maintenance plan + service contracts.
- Annual review & reporting to ARS.
- Stakeholder map (ARS, préfecture, Agence de l'Eau, hygiène).

EcoDialysis delivers **both** stacks as a single repeatable playbook.

Sources: Gov S26 S23

Why every site needs a tailored playbook

A site-specific playbook is not optional. The same RO reject does not have the same destiny in every hospital.

WHAT VARIES BY SITE

- **Water quality:** mains hardness, chloride, silica — drives reuse polishing need.
- **Tariff structure:** water + sewer per m³ — drives ROI.
- **Layout:** dialysis unit floor, toilets distance, technical loops accessibility — drives CAPEX.
- **RO age:** determines reject volume (220–320 L).
- **Demand profile:** session schedule vs reuse-point demand — drives storage sizing.
- **Governance:** CHU vs CH vs private — drives sign-off path.

WHY A GENERIC TANK FAILS

- Wrong size → either overflow or starvation.
- Wrong polishing → biofilm or chemistry mismatch.
- Wrong network → backflow risk, ARS rejection.
- Wrong demand match → empty <30% of the time.
- Wrong reporting → no ROI verification, no certification.

Hospitals do not need another ESG dashboard.

They need an operational water-resilience playbook.

Part 3 — Proof

International centers already prove that reuse and optimization work.
The hard part is not the tank — it is the repeatable site-specific dossier.

International proof — eight reference centers

CANTERBURY · UK 800 L/h RO-reject recovery, ~£7,500/yr. Ope	BRISTOL · UK 13,140 m³/yr, >£15,000/yr. Ope	ASHFORD · UK Design-in route at ~£2,500. Ope	EAST KENT · UK Heat recovery 0.86 kWh/Tx, 27,905 kWh/yr. Ope Gem
BENDIGO · AU RO-reject reuse, named ANZSN benchmark.	QUEEN ELIZABETH · AU Modern RO retrofit, lower L+kWh per Tx.	BARWON · AU Process optimization & metering case.	SPAIN · 20 CENTERS Mean 476.5 L/session; daily-opening 430 L vs 3×/wk 630 L. Ope Gem

All evidence aggregated and verifiable via Appendix B (case-study table) and Appendix I (sources).

UK case cluster — Canterbury · Bristol · Ashford · East Kent

CENTER	INTERVENTION	VOLUME SAVED	MONEY SAVED	ENERGY / CO-BENEFIT	SOURCE
Canterbury	RO-reject recovery → toilets & cleaning	800 L/h	~£7,500/yr	—	Ope
Bristol	RO-reject reuse (large center)	13,140 m³/yr	>£15,000/yr	—	Ope
Ashford	Design-in at new build	n/a low capex	~£2,500 design-in	—	Ope
East Kent	Heat exchanger recovery	—	—	0.86 kWh/Tx · 27,905 kWh/yr	Ope Gem

PATTERN

- Design-in beats retrofit on capex (Ashford).
- Bigger centers compound savings (Bristol).
- RO-reject is the dominant lever (Canterbury, Bristol).
- Heat recovery is a second lever — pairs with reject reuse (East Kent).

TRANSLATION TO FRANCE

- UK uses NHS-style local approvals — France adds ARS / EICH layer.
- UK tariffs differ — French ROI uses local water+sewer per m³.
- UK shows reuse is technically routine — the missing layer in France is the dossier.

Australia / New Zealand — ANZSN, Bendigo, Queen Elizabeth, Barwon

ANZSN AGGREGATED IMPACT

- >160,000 L/yr water saved
- 6.4 MWh/yr electricity saved
- 720 L citric acid solution avoided
- ~AU\$2,400 product savings

Source: [Ope](#)

MODERN VS OLD RO — MEASURED

	OLD	MODERN
L / treatment	548	357
kWh / treatment	7.2	3.1

RO replacement alone yields a ~35% water + 57% energy per-treatment cut. [Ope](#)

WHY THIS MATTERS FOR THE PLAYBOOK

- Even **without** reuse, RO replacement is a major lever.
- Disinfection optimization (citric / heat) saves chemicals + water + carbon together.
- Process improvement and reuse are **additive**, not alternatives.
- ANZSN proves a green-nephrology network can publish results — basis for the EcoDialysis label.

BENDIGO · QUEEN ELIZABETH · BARWON

Each is a named ANZSN benchmark in RO-reject reuse, modern-RO retrofit, and process metering respectively. They form a triple-case template the French market currently lacks. [Ope](#) [Gem](#)

Process optimization proof — Spain, modern RO, disinfection

SPAIN · 20-CENTER BENCHMARK

COHORT	L / HD	KWH / HD
20-center mean	476.5	—
Daily-opening centers	430.20	10.82
3×/week centers	629.81	16.79

Schedule shape is a **direct** water + energy lever. Ope Gem

THE THREE OPTIMIZATION LEVERS

- **Equipment:** modern RO (357 L vs 548 L; 3.1 kWh vs 7.2 kWh). Ope
- **Schedule:** daily vs 3×/week pattern (430 L vs 630 L / HD). Ope
- **Disinfection:** citric vs heat, frequency optimization, water + chemical avoidance. Gem S26

INSIDE THE DIALYSATE LOOP

- Qd reduction (e.g. 500 → 400 ml/min): **~24 L dialysate** savings/session in cited framing. S26
- **60–100 L** pretreated water economy per session in same context. S26
- HDF tuning, conductivity-routed reject. Gem

French proof — Saint-Exupéry, AP-HM, SFNDT playbook

CLINIQUE SAINT-EXUPÉRY · TOULOUSE

- ~1,128 m³/yr water saved (RO-reject reuse). **S26**
- Investment ~€30k, payback 5–6 years.
- French operational template — the closest match to EcoDialysis reference center.

AP-HM · MARSEILLE

- Larger-reuse-potential studies referenced.
- Shows CHU-scale interest and ARS pathway feasibility. **S26**

SFNDT 2026 PLAYBOOK — WHAT IT CONFIRMS

- RO-reject reuse is a recommended action sheet. **S26**
- Centralized acid-concentrate reduces waste & consumption.
- Storage constraints: duration, temperature, distance.
- Separate-network requirements.
- No direct leap to high-risk reuse.
- Energy management obligations, waste reduction, staff actions, RSE communication.

France has the cases.

France has the guide.

France lacks the **repeatable dossier engine**.

Part 4 — Product

Not a tank. Not an ESG report. A site-specific water-resilience playbook delivered as audit + dossier + implementation + monitoring + label.

Why not just buy a tank?

THE TANK FANTASY

"Just install a tank under the RO and reuse the water."

This view dies on first contact with the hospital water plant.

WHAT A TANK CANNOT DO

- ✗ Cannot characterize the reject stream
- ✗ Cannot match volumes to non-clinical demand
- ✗ Cannot file the EICH dossier
- ✗ Cannot operate the separated network
- ✗ Cannot pass CLIN review
- ✗ Cannot prove the savings
- ✗ Cannot scale across a GHT

WHAT ECODIALYSIS DOES INSTEAD

- ✓ **Audit** — 4-point metering + bills + plans + RO inventory.
- ✓ **Map** — green/orange/red reuse routes for this exact site.
- ✓ **Size** — storage + network + treatment matched to demand profile.
- ✓ **File** — EICH-compliant ARS dossier with sign-off chain.
- ✓ **Implement** — partner with plumbers / integrators / RO vendors.
- ✓ **Meter** — continuous SaaS monitoring + monthly verification.
- ✓ **Certify** — EcoDialysis label per site, benchmarkable.

A tank is a component.

A **dossier engine** is the product.

Product architecture — audit · playbook · implementation · monitoring · label

1 · AUDIT

- Site visit + data pull
- 4-point meter install
- Bills & tariff analysis
- Demand map
- Baseline 4–6 wk

€8–18k

2 · PLAYBOOK

- Green / orange / red map
- Scenario ROI
- EICH dossier draft
- Risk register
- Go / no-go

€10–25k

3 · IMPLEMENTATION

- Tender support
- Integrator coordination
- Hygiene sign-off
- Commissioning
- Staff training

5–15 % of CAPEX

4 · MONITORING

- SaaS dashboard
- 4-point + conductivity
- Monthly verification
- ARS reporting
- Alarms & carnet

€600–1,400 / month

5 · LABEL

- EcoDialysis certification
- Benchmark database
- Annual audit
- Public registry
- Network effect

€3–8k / yr

Wedge = audit. Repeatability = playbook. Recurring = monitoring. Network = label. Defensibility = benchmark database. **Pch** **Per** **S26**

Equipment & supplier ecosystem

LAYER	EQUIPMENT	REFERENCE SUPPLIERS	ECODIALYSIS TOUCHPOINT
Dialysis generator	Hemodialysis machines	Fresenius, B. Braun, Baxter/Vantive, Nikkiso, Nipro, JMS, Quanta, Outset/Tablo, NxStage	Qd / HDF tuning, energy submetering
Water-treatment plant	Pretreatment + RO skid + loop	B. Braun AQUAboss, Fresenius, integrators	4-point metering, reject characterization
Dialyzer + bloodlines	Membranes & circuits	Asahi Kasei, Toray, Fresenius, B. Braun, Nipro	Procurement scorecard (Phase 3)
Acid concentrate / bicarbonate	Liquid + cartridge	Fresenius, B. Braun, Baxter, regional	Centralized distribution playbook S26
Disinfection	Citric, heat, peracetic	Vendor-bundled	Frequency optimization Gem Ope
Reuse hardware	Tanks, pumps, UV-C, backflow prevention	Local integrators	Sizing, anti-backflow class (EA/EB/AA)
Metering & IoT	Flow, conductivity, energy	Endress+Hauser, Siemens, regional	SaaS data ingestion
Waste & consumables	DASRI/DAOM	Veolia, Suez, regional	Sorting + reduction (Phase 3)
Future / pilot	NeoKidney, AWAK, REDY sorbent	NextKidney, AWAK, vendor R&D	Watch & pilot integration Gem

Detailed supplier landscape with caveats & partnership roles in Appendix A11.

Technology maturity — ready now / site-specific / pilot only

LEVER	MATURITY	WATER	ENERGY	WASTE	CLINICAL RISK	ECODIALYSIS PHASE
RO-reject → non-clinical reuse	Ready	+++	0	+	None (boundary)	Phase 1
RO replacement (357 L vs 548 L)	Ready	+++	+++	+	None	Phase 1
Disinfection optimization	Ready	++	+	++	Vendor sign-off	Phase 1
Heat recovery (East Kent)	Ready	0	++	0	None	Phase 2
Qd reduction (500 → 400 ml/min)	Site-specific	++	+	+	Clinical sign-off	Phase 2
Centralized acid concentrate	Site-specific	+	+	++	Low	Phase 2
Conductivity-routed reuse	Site-specific	++	0	0	Engineering	Phase 1–2
Pretreatment recovery RO	Site-specific	++	–	0	Vendor	Phase 2
HDF dialysate volume tuning	Site-specific	+	0	0	Clinical sign-off	Phase 2
Sorbent / portable (REDY, AWAK)	Pilot only	+++	+	+	High	Watch
NeoKidney wearable	Pilot only	+++	+	+	High	Watch
Spent dialysate reuse	Not deployable	n/a	n/a	n/a	Boundary	Never

Full 18-row matrix in Appendix A13. Sources: [Gem](#) [Ope](#) [S26](#)

Business model — audit · playbook · SaaS · success · label

FIVE REVENUE LINES

LINE	LOGIC	RANGE
Audit	Wedge, paid pilot	€8–18k
Playbook + EICH dossier	Per site	€10–25k
Implementation support	% of CAPEX	5–15 %
SaaS monitoring	Recurring	€600–1,400 / mo
Success fee	% of verified savings	10–20 % yr 1–3
Label + benchmark	Annual	€3–8k / yr

Pricing illustrative; calibrated on first pilots.

CHANNELS & EXPANSION

- **Direct:** CHU sustainability office, biomedical engineering.
- **GHT rollout:** one CH wins → cluster follows.
- **Private networks:** AURA / Diaverum / NephroCare / Echo-style chains.
- **Supplier partnerships:** RO & generator vendors include EcoDialysis audit at sale.
- **Subsidy partners:** Agences de l'Eau (basin-specific), CEE (verify category), ADEME.

WHY THIS COMPOUNDS

- Audit data → benchmark database → defensibility.
- Playbook reuse → margin expansion.
- Label adoption → network effect among nephrology directors.

Buyer map — who signs, who pays, who scales

BUYER	PAIN	DECISION UNIT	WEDGE	EXPANSION
CHU · large public	Reporting + reputation + scarcity	Sustainability office + bioméd + direction	Audit + flagship pilot	Other departments (sterilization, laundry)
CH / GHT	Multi-site scarcity + opex	GHT direction + technique	One CH pilot → GHT framework	Cluster rollout
Private dialysis network	Margin compression + ESG mandates	Medical director + ops + finance	Network-wide audit deal	Cross-country rollout
Medium independent unit	Opex sensitivity	Owner-medical-director	Audit + light SaaS	Reference for similar units
New-build / renovation	Design-in window	Architect + bioméd + procurement	Ashford-style design-in	Architect partnerships

Reference market: ~2,965 health establishments · ~1,100–1,200 dialysis units · 51,700 patients · 77 % in-center/UDM. **Per**

Part 5 — Pilot, risk & ask

Three pilots. Six months. A repeatable dossier engine for France.

Three-pilot plan — CHU · GHT · private network

PILOT A · CHU FLAGSHIP

Target: AP-HM / CHU Toulouse / CHU Lyon class.

- ~150 patients
- Full reuse + heat recovery
- EICH dossier proof
- SFNDT publication path

6 months

€80–120k

PILOT B · GHT MID-SIZE

Target: CH in water-stressed basin (Adour-Garonne / Loire).

- ~80 patients
- Reuse + RO upgrade scoping
- GHT framework agreement
- Agence de l'Eau co-funding

6 months

€60–90k

PILOT C · PRIVATE NETWORK

Target: AURA / Diaverum / regional chain.

- 3 sites, ~100 patients each
- Repeatable playbook test
- SaaS + label deployment
- Network-rollout option

9 months

€140–200k

All three together test: technical feasibility, dossier repeatability, and channel scalability.

What we need from each pilot hospital

ACCESS

- Water plant & RO room access
- Technical plans (water, drainage, electricity)
- Dialysis floor session schedule
- Non-clinical demand map (toilets, cleaning, HVAC, garden)
- Permission to install 4 meters

DATA

- 12 months water + sewer bills
- RO + generator inventory + age
- Energy bills for water-treatment area
- Session log (sessions/yr per machine)
- Existing maintenance & disinfection logs

PEOPLE

- Bioméd / services techniques referent
- Hygiene / CLIN referent
- Direction sponsor
- Medical referent (nephrology)
- ARS / préfecture pre-file contact

No clinical data. No patient data. The hospital remains the data controller and operational owner.

What we deliver in 30 days

DAY 1 – 10 · ECO-AUDIT

- Site visit + interviews
- 4-point meter install
- Bills + plans analysis
- RO + generator inventory

DAY 10 – 20 · MAPPING

- Green/orange/red reuse routes
- Demand profile vs reject profile
- Three reuse scenarios sized

DAY 20 – 25 · SCENARIO ROI

- Low / central / high reject
- 50 / 70 / 90 % capture
- Tariff + CAPEX sensitivity
- Subsidy stacking option

DAY 25 – 30 · DECISION PACK

- Risk register
- EICH dossier draft
- Go / no-go recommendation
- Implementation timeline

Output: A4 dossier + 1-page board memo + live SaaS dashboard baseline.

What we refuse to do

BOUNDARIES WE WILL NOT CROSS

- ✗ No patient-contact reuse, ever.
- ✗ No reuse fed back to dialysis-water production.
- ✗ No spent-dialysate reuse.
- ✗ No guarantee of ARS approval — every dossier is local.
- ✗ No unverified CNAM / tariff claim.
- ✗ No full carbon-neutrality claim — water adaptation first.
- ✗ No black-box calculation — methodology is published.
- ✗ No vendor lock-in recommendation without comparative evidence.

Trust is the wedge.

What we refuse to do is the moat.

HOW THIS MAPS TO CLAIM CONTROL

- Methodology box on every report.
- Objective, perimeter, assumptions, uncertainty. **TSP**
- Caveat language standardized.
- Forbidden-claim list maintained in Appendix A16.

KPIs & success criteria

$\geq 50 \%$

REJECT CAPTURE RATE
Target after commissioning

$\geq 70 \%$

SUBSTITUTION RATE
Non-clinical demand met

$< 6 \text{ yr}$

PAYBACK
Saint-Exupéry benchmark

0

CLINICAL INCIDENTS
Boundary held

OPERATIONAL KPIS

- L/session reject — measured continuously.
- L/session substituted — measured at reuse meter.
- kWh/session — RO + dialysate loop.
- Disinfection chemical avoided.
- Conductivity excursions = 0.

DOSSIER & TRUST KPIS

- Time-to-dossier (target ≤ 60 days).
- ARS approval rate.
- Carnet sanitaire compliance.
- Monthly verification on time.
- EcoDialysis-labelled centers count.

Risk & trust — what could go wrong, how we protect

RISK	WHY IT MATTERS	MITIGATION	OWNER
ARS rejection / delay	No deployment without dossier	Pre-file early; templated EICH dossier	EcoDialysis + hospital direction
Biofilm / cross-connection	Patient & staff safety	Separated network, anti-backflow (EA/EB/AA), monitoring, emergency stop	Bioméd + hygiène
Demand-supply mismatch	Empty reuse loop = no ROI	Pre-sizing storage + demand mapping	EcoDialysis
Tariff weakening ROI	Low water/sewer tariff = poor payback	Site triage; refuse low-ROI sites; subsidy stacking	EcoDialysis
Capex retrofit cost	Layout-dependent	Site triage; design-in for new-build (Ashford)	Hospital + EcoDialysis
Overclaiming ESG / carbon	Loss of trust	Methodology box; refusal list; published caveats	EcoDialysis
Patient / staff perception	Social licence	Communication kit + boundary clarity	Hospital direction + EcoDialysis
Supplier lock-in	Bias risk	Comparative evidence requirement, vendor-neutral specs	EcoDialysis

Full register with evidence sources in Appendix A14.

Defensibility — five compounding moats

BENCHMARK DB

Every audit feeds a France-specific water/energy benchmark unavailable elsewhere.

PLAYBOOK LIBRARY

Repeatable EICH dossiers shorten each new project — margin compounds.

SOURCE-LABELLED EVIDENCE

Every claim chip-linked to a primary source — defensible vs consultants.

ECODIALYSIS LABEL

Network effect among nephrology directors and ARS reviewers.

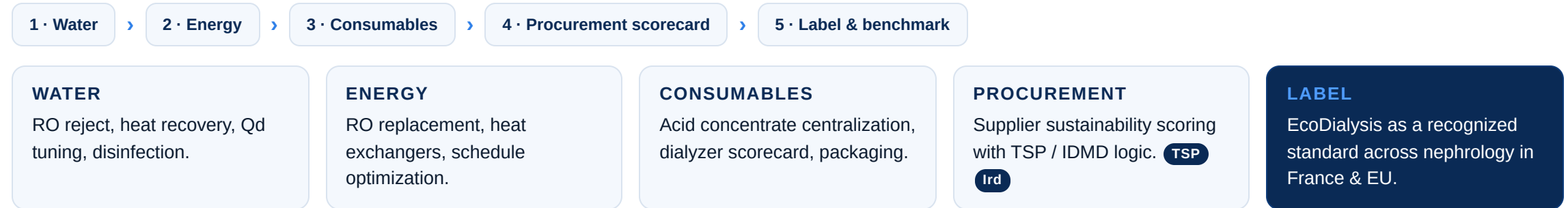
SUPPLIER SCORECARD

Procurement leverage as we expand into devices and consumables.

EcoDialysis wins when every dialysis unit can say:

we know our water · we know our safe reuse route · we can prove the savings.

Future expansion — water first, then four more layers



Hospitals do not need another ESG dashboard.

They need an operational water-resilience playbook.

EcoDialysis builds it — site by site, dossier by dossier, certification by certification.

Three pilots · six months · one repeatable engine for France.

Adaptation first · mitigation second · boundary non-negotiable

Appendix — diligence evidence library

23 extracted sources · 0 extraction failures · all key facts traced to a primary document.

A1 · Source map — 23 extracted sources, zero extraction failures

CHIP	ID	SOURCE	TYPE	QUALITY
Per	DR-France	EcoDialysis_Perplexity.pdf (+ .docx)	Deep-research strategy synthesis	Medium
Gem	DR-Tech	Clinical and Technical Advancements in Dialysis_Gemini.pdf (+ .docx)	Deep technical synthesis	Medium-high
Ope	DR-Water	Water Sustainability Practices in Dialysis_OpenAI.pdf	Deep-research synthesis	Medium-high
S23	SFNDT-2023	SFNDT_guide complet-VF-HD-2_260526.pdf	Professional society guidance	High
S26	SFNDT-2026-L	Guide_des_bonnes_pratiques_en_néphrologie_verte_Ve_260526.pdf	Professional society guidance (long)	High
S26	SFNDT-2026-S	Guide_de_la_néphrologie_verte_Version_courte_2026_260526.pdf	Professional society guidance (short)	High
Gov	Gov-Health-Roadmap	Planification écologique du système de santé.pdf	Government roadmap	High
TSP	TSP-Health	TSP-Decarbonons-les-industries-des-medicaments-RF.pdf	The Shift Project report	High
TSP	TSP-Pathway	[PPJS] Guide méthodologique pour l'évaluation carbone des parcours de soin.pdf	Methodological guide	High
TSP	TSP-Med	240404-Rapport-Decarbonons-l'Autonomie-The-Shift-Project-1-1.pdf	Report	Medium-high
Ird	IRDES-Sustainability	IRDES · Soutenabilité environnementale des systèmes de santé.pdf	Literature review	High
HCW	HCWH	L'empreinte climatique du secteur de la santé Health Care Without Harm.pdf	NGO global analysis	Medium-high
HCW	Global-Green-Health	Green health how to decarbonise global healthcare systems.pdf	Global report	Medium-high
HCW	Climate-friendly-healthcare	Climate-friendly healthcare reducing the impacts of the healthcare sector.pdf	Global report	Medium
Pch	Pitch-v3	EcoDialysis_Pitch_v3.pptx + .pdf	Internal pitch	High (narrative)
Pch	Base-v1	EcoDialysis Pitch_v1.html	Internal HTML baseline	Internal style
—	Vendor-block	Official corporate / regulatory pages (Fresenius, B. Braun, Vantive/Baxter, Nipro, Nikkiso, Asahi Kasei, Toray, JMS, Quanta, NxStage, SNITEM/C2DS/AFNOR SPEC 2313)	Corporate & regulatory	Medium-high to High

Reliability legend — High: official regulation, government, professional society. Medium-high: official corporate, peer-reviewed. Medium: secondary synthesis. Low: excluded.

A2 · Source chip registry

PER

France market & GTM deep research (Perplexity).

Per

GEM

Technology deep research (Gemini).

Gem

OPE

Water sustainability deep research (OpenAI).

Ope

S23

SFNDT 2023 green nephrology guide.

S23

S26

SFNDT 2026 long & short guides.

S26

GOV

Planification écologique du système de santé.

Gov

TSP

The Shift Project (health, pathway, autonomy).

TSP

IRD

IRDES literature review.

Ird

HCW

Health Care Without Harm + global green health.

HCW

PCH

EcoDialysis pitch v1 / v3 baseline.

Pch

A3 · DR-Water insight table — 18 facts

#	FACT	QUANTITY
1	Canterbury RO-reject recovery	800 L/h
2	Canterbury annual savings	~£7,500/yr
3	Ashford design-in route	~£2,500
4	Bristol water saving	13,140 m³/yr
5	Bristol annual savings	>£15,000/yr
6	East Kent heat exchangers	0.86 kWh/treatment
7	East Kent projected annual recovery	27,905 kWh/yr
8	ANZSN aggregated water saved	>160,000 L/yr
9	ANZSN aggregated electricity saved	6.4 MWh/yr
10	ANZSN citric acid avoided	720 L
11	ANZSN product savings	~AU\$2,400
12	Modern AU RO water per treatment	357 L (vs 548 L old)
13	Modern AU RO energy per treatment	3.1 kWh (vs 7.2 kWh old)
14	Spain 20-center mean	476.5 L/session
15	Spain daily-opening centers	430.20 L/HD · 10.82 kWh/HD
16	Spain 3×/week centers	629.81 L/HD · 16.79 kWh/HD
17	Named benchmarks	Bendigo · Barwon · Queen Elizabeth
18	Future / pilot track	REDY · NeoKidney · AWAK

Primary chip: **Ope** (DR-Water). Cross-confirmed by **Gem**.

A4 · DR-France insight table — 12 facts

#	FACT	VALUE
1	French dialysis patients	~51,700
2	In-center / UDM share	~77 %
3	Feedwater per session	~400–500 L
4	RO reject fraction	50–70 %
5	First adopters	CHU / CHRU + large private
6	Health establishments	~2,965 (context-dependent)
7	Dialysis units (est.)	~1,100–1,200
8	Reference center	100 patients
9	Sessions / patient / yr	156
10	Total sessions / yr	15,600
11	Reject / session assumption	270 L (low 220 · high 320)
12	Reference reject volume	~4,200 m³/yr · ~1.0–1.1 tCO₂e water-only caveat

Primary chip: **Per** . Carbon caveat aligned with **TSP** methodology.

A5 · DR-Tech insight table — 15 facts

#	FACT
1	Taxonomy: reuse · RO optimization · disinfection · heat recovery
2	Supports 476.5 L/session multicenter benchmark
3	High-recovery RO architecture pathway
4	Conductivity-based routing logic for reuse decisions
5	Consolidates Canterbury / Ashford / Bristol evidence
6	Confirms ANZSN disinfection optimization logic
7	Heat-recovery cases (East Kent, Newcastle)
8	Distinguishes low-risk non-clinical vs higher-risk pathways
9	Sorbent / portable systems as emerging track
10	NeoKidney and AWAK pilot status
11	Retrofit cost and governance barriers identified
12	Recommends staged implementation roadmap
13	Reinforces metering and process controls
14	Positions spent dialysate reuse as not first-wave
15	Supports evidence-level stratification table

Primary chip: **Gem** . Reinforces maturity matrix in Appendix A13.

A6 · SFNDT 2023 — green nephrology framework

#	INSIGHT
1	Structured green nephrology action framework
2	Water · energy · waste levers integrated
3	Governance and team-level ownership emphasized
4	Training and implementation culture central
5	Procurement and life-cycle orientation
6	Risk-aware operational transition framing
7	Dialysis footprint reduction hierarchy
8	Measurement-first logic (matches EcoDialysis architecture)
9	Continuous improvement approach
10	Quality-and-safety co-framing of sustainability
11	Multi-domain indicator tracking
12	Basis for actionable checklists

Primary chip: **S23** . Clinical credibility anchor.

A7 · SFNDT 2026 long guide — 20 operational facts

#	FACT	QUANTITY IF ANY
1	Detailed action sheets for dialysis & nephrology operations	—
2	Qd reduction pathway with quantified per-session savings	see #3, #4
3	Dialysate savings from 500 → 400 ml/min framing	~24 L/session
4	Pretreated water economy in the same context	~60–100 L/session
5	Multicenter French Qd evaluation context	—
6	HDF optimization & dialysate volume tradeoffs	—
7	Centralized acid-concentrate distribution	waste + consumption ↓
8	Dedicated section on RO-reject reuse	—
9	French case: Clinique Saint-Exupéry Toulouse	—
10	Saint-Exupéry reported saving	~1,128 m³/yr
11	Saint-Exupéry investment / payback	~€30k · 5–6 yr
12	AP-HM Marseille larger-reuse-potential studies	—
13	Practical storage constraints (duration, temp, distance)	—
14	Separate-network requirements	—
15	No direct leap to high-risk reuse	boundary
16	Energy management obligations	—
17	Waste reduction pathways	—
18	Staff / organizational actions	—
19	RSE and communication logic	—
20	Reusable as implementation playbook	—

Primary chip: **S26** . The closest match to EcoDialysis operational playbook.

A8 · Government / regulation appendix — EICH & Décret 2024-796

DÉCRET N°2024-796 · IN FORCE 1 SEPT 2024

- EICH framework = Eaux Impropres à la Consommation Humaine.
- Covers sensitive establishments including healthcare.
- Authorizes non-clinical reuse subject to ARS / préfecture file.
- Mandatory separated network.
- Anti-backflow devices required.
- Signage program required.
- Carnet sanitaire entries required.
- Monitoring & maintenance plan required.
- Emergency stop to potable fallback required.
- No patient-contact reuse permitted.

STAKEHOLDER MAP

- **ARS** — health authority sign-off.
- **Préfecture** — administrative authorization where required.
- **Agence de l'Eau** — co-funding by basin.
- **CLIN / hygiène** — internal safety committee.
- **Direction** — administrative sponsor.
- **Services techniques + bioméd** — operational owners.
- **Médical + nephrology** — clinical sign-off.

Remaining uncertainties (carried into Appendix A16): exact ARS pathway per basin · monitoring frequency · tariff structure interaction · Agence de l'Eau eligibility · CEE category. Sources: **Gov** **S26**

A9 · Health-system decarbonization context — why water is adaptation first

MACRO CONTEXT

- Healthcare > 8 % of national GHG footprint. **Gov** **HCW**
- ~50 MtCO₂e order of magnitude.
- Offer of care ~45 % · medicines & medical products ~55 %.
- Scope 3 procurement dominates.

HEALTH-SYSTEM ROADMAP AXES

- Buildings & energy
- Industries / products
- Sustainable procurement
- Eco-responsible care
- Waste · transport · digital · training

Source: **Gov** Planification écologique.

WHY WATER IS ADAPTATION FIRST

- Water resilience is a continuity issue, not an ESG metric.
- Tariff & scarcity hit opex immediately.
- Carbon weight of water is real but secondary (1.0–1.1 tCO₂e on 4,200 m³). **Per**
- **TSP**
- Medical devices & consumables matter for Scope 3 — addressed in EcoDialysis Phase 3.
- System-level resilience framing aligns with TSP & IRDES. **TSP** **Ird**

A10 · Equipment ecosystem — environmental, data, intervention, boundary

LAYER	ENVIRONMENTAL ISSUE	DATA NEEDED	INTERVENTION	BOUNDARY	OWNER
Dialysis generator	Energy + water tuning	Sessions/machine, Qd settings	Qd reduction, HDF tuning	Clinical sign-off	Bioméd + nephro
Water-treatment plant	Reject volume	Mains-in, permeate, reject flows	Modern RO, conductivity routing	Vendor sign-off	Services techniques
RO skid	L+kWh / treatment	Age, model, performance	Replacement / retrofit	Vendor	Bioméd
Pretreatment	Chemical use, backwash	Filter change cycles	Optimization, recovery RO	Vendor	Bioméd
Dialysate loop	Disinfection cycles	Citric / heat frequency	Optimization, chemical reduction	Vendor sign-off	Bioméd + hygiène
Dialyzer membranes	Single-use waste	Vendor scorecard	Procurement scoring (Phase 3)	Clinical	Direction + médical
Bloodlines / needles / catheters	Single-use waste	Volume + supplier	Packaging + procurement	Clinical	Pharma + procurement
Acid concentrate	Logistics + packaging	Local vs central feed	Centralized distribution S26	Hygiène	Services techniques
Bicarbonate	Cartridge waste	Consumption / session	Procurement scorecard	Clinical	Pharma
Saline / disinfectants	Chemicals	Volumes	Optimization	Hygiène	Pharma + hygiène
Packaging	Cardboard / plastic	Supplier audit	Procurement scorecard	Logistics	Procurement
DASRI / DAOM waste	High-carbon waste stream	Sort rates	Sorting + reduction Ird	Hygiène	Services généraux
Sensors / meters	—	Existing inventory	4-point install	—	EcoDialysis
Tanks / pumps	Reuse hardware	Demand profile	Sizing	Engineering	Integrator
UV-C	Reuse polishing	Flow + microbio	Inline reuse loop	Vendor	Integrator
Backflow prevention	Cross-connection risk	Network plan	EA/EB/AA per use	Regulatory	Plumbing + hygiène

A11 · Supplier landscape (verify local relevance)

COMPANY	CATEGORY	RELEVANT PRODUCTS	ECODIALYSIS LEVER	CAVEAT
Fresenius Medical Care	Generator + RO + dialyzer	5008/6008, AquaA/AquaB, dialyzers, NxStage	Modern RO, generator tuning, procurement	Vendor neutrality required
B. Braun	Generator + water	Dialog, AQUAboss	Water plant, reject reuse	—
Baxter / Vantive	Generator + PD	Artis, AK series; PD	Generator tuning	Recent spin-off
Nipro	Generator + dialyzer	Surdial, dialyzers	Procurement scorecard	EU footprint variable
Nikkiso	Generator	DBB-EXA, water systems	Generator tuning	—
Asahi Kasei Medical	Dialyzer	Membranes	Procurement	—
Toray	Dialyzer	Membranes	Procurement	—
JMS	Dialyzer + bloodlines	Products	Procurement	Regional footprint
Quanta	Generator	SC+ portable	New low-water entrant	Pilot scale
Outset / Tablo	Generator	Tablo	Water-light alternative	US-focused
NextKidney (NeoKidney)	Future	Wearable	Watch & pilot	Pre-deployment
AWAK	Future	Sorbent	Watch & pilot	Pre-deployment
Endress+Hauser / Siemens	Metering / IoT	Flow, conductivity	SaaS data layer	—
Veolia / Suez	Water & waste	Operator	Reuse hardware, waste audit	Bias risk → vendor neutrality
SNITEM / C2DS / AFNOR SPEC 2313	Procurement framework	IDMD	Scorecard alignment	—

Sources: official corporate pages + vendor landscape evidence block (see A1).

A12 · Financial model — scenarios & sensitivities

REJECT SCENARIO × CAPTURE RATE

REJECT	M ³ /YR	50 % CAPTURE	70 %	90 %
Low 220 L	3,432	1,716	2,402	3,089
Central 270 L	4,212	2,106	2,948	3,791
High 320 L	4,992	2,496	3,494	4,493

All figures m³/yr for the reference 100-patient center. **Per**

WHAT CANNOT BE QUANTIFIED YET

- Exact CEE category and €/MWh equivalence for water reuse.
- Agence de l'Eau co-funding per basin and project.
- CNAM tariff impact (none assumed — caveat required).
- Carbon credit eligibility (not claimed).

TARIFF & PAYBACK SENSITIVITY

WATER + SEWER €/M ³	SAVINGS AT 2,948 M ³ REUSE	INDICATIVE PAYBACK @ €30K
2.50	~€7,400/yr	~4 yr
3.50	~€10,300/yr	~3 yr
5.00	~€14,700/yr	~2 yr

Saint-Exupéry anchor: ~1,128 m³, ~€30k, 5–6 yr payback. **S26**

DECISION RULE

- Refuse sites with tariff < €1.80/m³ unless subsidy or scarcity-risk override.
- Prefer design-in over retrofit when window exists (Ashford pattern).
- Couple reuse with RO replacement when RO > 10 yr.

A13 · Technology maturity full matrix

LEVER	MATURITY	EVIDENCE	WATER	ENERGY	WASTE	CAPEX	REG	PHASE
RO-reject non-clinical reuse	Ready	UK + AU + ES	+++	0	+	Med	EICH dossier	1
Modern RO replacement	Ready	AU measured	+++	+++	+	High	Vendor	1
Heat recovery (exchangers)	Ready	East Kent	0	++	0	Med	Eng	2
Disinfection optimization	Ready	ANZSN	++	+	++	Low	Vendor	1
Conductivity-routed reuse	Site	Tech synthesis	++	0	0	Low	Eng	1–2
Pretreatment recovery RO	Site	Tech synthesis	++	–	0	Med	Vendor	2
Qd reduction (500 – 400)	Site	SFNDT 2026	++	+	+	Low	Clinical	2
HDF dialysate volume tuning	Site	SFNDT 2026	+	0	0	Low	Clinical	2
Centralized acid concentrate	Site	SFNDT 2026	+	+	++	Med	Hygiène	2
Schedule shape (daily vs 3x/wk)	Site	Spain	+++	+++	0	Low	Clinical	2
Toilet flushing reuse	Ready	UK	++	0	0	Low	EICH	1
Cleaning & outdoor reuse	Ready	UK	+	0	0	Low	EICH	1
HVAC cooling-tower make-up	Site	Industry	++	0	0	Med	Eng	2
Laundry pre-wash reuse	Site	Cross-sector	++	0	0	Med	EICH	2
Sterilization feed (non-final)	Site	Cross-sector	+	0	0	Med	EICH + hygiène	2
Sorbent / portable (REDY/AWAK)	Pilot	Vendor R&D	+++	+	+	—	High	Watch
NeoKidney wearable	Pilot	Vendor R&D	+++	+	+	—	High	Watch
Spent dialysate reuse	Boundary	—	n/a	n/a	n/a	—	Forbidden	Never

A14 · Risk register

RISK	WHY	MITIGATION	OWNER	EVIDENCE
ARS rejection / delay	No reuse without dossier	Pre-file, templated EICH dossier, prior-art casework	EcoDialysis + direction	Gov S26
Biofilm / infection	Patient safety	UV-C polishing, anti-stagnation, monitoring	Bioméd + hygiène	S26 Gem
Cross-connection	Potable contamination	Separated network, EA/EB/AA backflow, signage	Plumbing + hygiène	Gov
Conductivity excursion	Reuse quality drift	Inline routing to drain	Bioméd	Gem
Demand-supply mismatch	Empty reuse loop	Storage sizing, demand mapping	EcoDialysis	S26 Ope
Retrofit capex blow-out	ROI risk	Site triage, design-in option (Ashford)	Direction + EcoDialysis	Ope
Low tariff weakens ROI	Payback > 7 yr	Site triage, subsidy stacking	EcoDialysis	Per
Procurement delays	Months added	Framework agreements, GHT route	Direction	—
Overclaiming ESG / carbon	Reputational	Methodology box, claim-control list	EcoDialysis	TSP
Patient / staff perception	Social licence	Communication kit, boundary clarity	Direction	S26
Ownership split	Operational drift	RACI per site, contracts	EcoDialysis	—
Supplier lock-in	Bias risk	Vendor-neutral specs, comparative evidence	EcoDialysis	TSP

A15 · Pilot implementation appendix — week-by-week

WEEK	PHASE	DELIVERABLES	HOSPITAL DATA NEEDED	APPROVALS
0	Mobilization	Kick-off, NDA, sponsor letter	Org chart, contacts	Direction sign-off
1–2	Site audit	Visit, plans review, RO inventory	Plans, RO logs, 12-mo bills	Services techniques
2–3	Metering install	4 meters + conductivity	Plumbing access	Bioméd + hygiène
3–7	Baseline	Continuous logging, demand mapping	Session schedule	—
7–9	Scenario design	Green/orange/red map, sizing, ROI	Tariffs	Direction + technique
9–12	Dossier draft	EICH file, risk analysis, sign-off chain	—	Hygiène + direction
12–14	Pre-file ARS	Informal review	—	ARS
14–18	Implementation	Plumbing, tank, UV-C, backflow, SaaS	Site access	CLIN, ARS final
18–20	Commissioning	Testing, training, signage	Staff time	Bioméd + hygiène
20–24	Verification	Monthly reports, KPI bundle	—	Direction + finance

RACI matrix and reporting templates available in playbook pack.

A16 · Claim control — allowed vs forbidden

ALLOWED CLAIMS

- ✓ "Captured X m³ of RO reject this month (M3 meter)."
- ✓ "Substituted Y m³ in non-clinical uses (M4 meter)."
- ✓ "Saved €Z based on local tariff bill verification."
- ✓ "Reduced disinfection cycles by N — chemicals avoided."
- ✓ "EICH-compliant dossier filed at ARS [date]."
- ✓ "Carnet sanitaire updated; zero conductivity excursions."
- ✓ "Energy savings X kWh from RO replacement (measured)."

FORBIDDEN CLAIMS

- ✗ "Carbon neutral" — never.
- ✗ "Zero impact" — never.
- ✗ "ARS-approved at every site" — local decision.
- ✗ "Patient water reused" — never.
- ✗ "Spent dialysate recycled" — never.
- ✗ "Equivalent to X cars off the road" without TSP methodology disclosure.
- ✗ "Eligible for CEE / Agence de l'Eau" without written confirmation.
- ✗ "Saves the hospital N €" without verified bill comparison.

Methodology box mandatory on every report: objective, perimeter, assumptions, uncertainty, data quality, caveats. **TSP**

Appendix I · Full source registry (chip anchors)

Per DR-France · Perplexity

EcoDialysis_Perplexity.pdf (+ .docx) — France market, ~51,700 patients, 77% in-center/UDM, reference-center model. Quality: Medium (secondary synthesis).

Gem DR-Tech · Gemini

Clinical and Technical Advancements in Dialysis_Gemini.pdf (+ .docx) — technology taxonomy, conductivity routing, maturity matrix support. Quality: Medium-high.

Ope DR-Water · OpenAI

Water Sustainability Practices in Dialysis_OpenAI.pdf — Canterbury, Bristol, Ashford, East Kent, ANZSN, modern vs old AU RO, Spain 20-center. Quality: Medium-high.

S23 SFNDT 2023

SFNDT_guide complet-VF-HD-2_260526.pdf — green nephrology framework, measurement-first, governance. Quality: High.

S26 SFNDT 2026

Guide_des_bonnes_pratiques_en_néphrologie_verte_Ve_260526.pdf + Guide_de_la_néphrologie_verte_Version_courte_2026_260526.pdf — Qd savings, Saint-Exupéry case, AP-HM, RO-reject reuse playbook. Quality: High.

Gov Government / regulation

Planification écologique du système de santé.pdf + Décret n°2024-796 (12 July 2024, in force 1 Sept 2024, EICH). Quality: High.

TSP The Shift Project

TSP-Decarbonons-les-industries-des-medicaments-RF.pdf · [PPJS] Guide méthodologique pour l'évaluation carbone des parcours de soin.pdf · 240404-Rapport-Decarbonons-l'Autonomie-The-Shift-Project-1-1.pdf. Quality: High.

Ird IRDES

La soutenabilité environnementale des systèmes de santé. Une revue de littérature sur l'empreinte écologique des systèmes de santé. Quality: High.

HCW Health Care Without Harm + Global Green Health

L'empreinte climatique du secteur de la santé Health Care Without Harm.pdf + Green health how to decarbonise global healthcare systems.pdf + Climate-friendly healthcare reducing the impacts of the healthcare sector on the world's climate.pdf. Quality: Medium-high.

Pch Pitch baseline

EcoDialysis_Pitch_v3.pptx + EcoDialysis-Pitch_v3.pdf + EcoDialysis Pitch_v1.html — internal narrative, positioning, GTM skeleton. Quality: High (narrative), internal style baseline.